



4.4.2 Via Gram-Schmidt QR factorization ¶

In [Section 3.2](#), you were introduced to the (Classical and Modified) Gram-Schmidt process and how it was equivalent to computing a QR factorization of the matrix, A , that has as columns the linearly independent vectors being orthonormalized. The resulting Q and R can be used to solve the linear least squares problem by first computing $y = Q^H b$ and next solving $R\hat{x} = y$.

Starting with $A \in \mathbb{C}^{m \times n}$ let's explicitly state the steps required to solve the LLS problem via either CGS or MGS and analyze the cost.:

- From [Homework 3.2.6.1](#) or [Homework 3.2.6.2](#), factoring $A = QR$ via CGS or MGS costs, approximately, $2mn^2$ flops.
- Compute $y = Q^H b$: $2mn$ flops.
- Solve $R\hat{x} = y$: n^2 flops.

Total: $2mn^2 + 2mn + n^2$ flops.